Designing drug-binding proteins with NISE

Zero-shot design of drug-binding proteins via neural iterative selection–expansion

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The question: can computation deliver tight binders directly?

Small-molecule recognition couples three design problems:

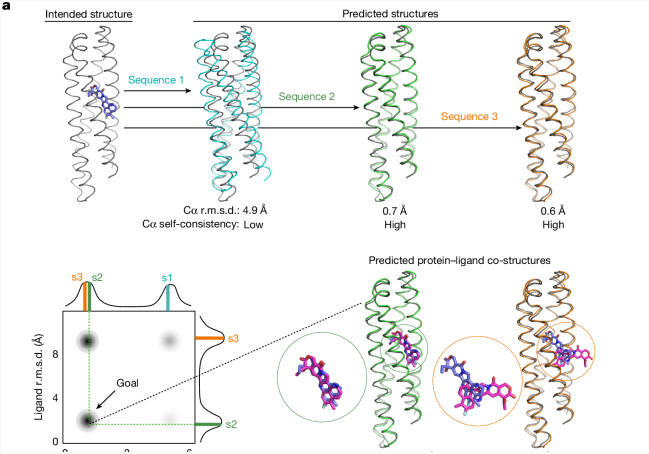
1. Choose a sequence that folds into a stable pocket.
2. Adjust the backbone to pack the ligand and support interactions.
3. Find a ligand position and conformation compatible with that pocket.

NISE alternates sequence design and protein–ligand structure prediction, selecting the best compatible pairs.

Here, **zero-shot** means no experimental screening or measured affinity guides the design trajectory. It still requires trained models, initial scaffolds, extensive computation and final experiments.

Main text: introduction, NISE sampling algorithm and Discussion.

A protein can fold correctly and still miss the ligand



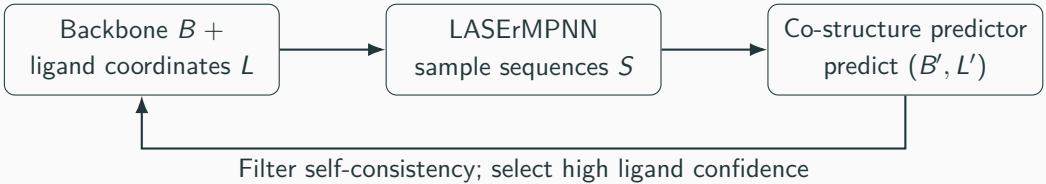
Self-consistency: agreement between the input design and the predicted complex.

Low backbone RMSD alone cannot establish the intended binding mode.

NISE also requires low ligand RMSD after protein alignment; ligand confidence then prioritizes candidates.

Fig. 1a, cropped; RMSD = root mean square deviation. Author schematic, CC BY 4.0.

Two reciprocal models define the design loop



sequence design: $P(S \mid B, L)$

complex prediction: $P(B, L \mid S, \text{ligand identity})$

Output: ranked sequences and their predicted drug-bound structures.

Adapted schematic from Fig. 1b,d. Joint-distribution optimization is the authors' interpretation, not a binding-energy guarantee.

A concrete iteration: exatecan in a four-helix pocket

1. Start with a backbone and docked exatecan; discard the starting sequence.
2. For each current pose, sample 1,000 complete sequences with LASErMPNN.
3. Predict each complex using RoseTTAFold All-Atom (RFAA). Keep protein confidence > 0.8 and backbone/ligand RMSD < 1.5 Å relative to that round's input.
4. Rank survivors by ligand pLDDT (local structure confidence); carry forward the best new poses.
5. Repeat for 35 iterations. Pool candidates for more stringent final filtering.

The original pose stays in the exatecan pool, alongside up to two improved poses. Ligand RMSD here uses ring-system heavy atoms, excluding the rotatable chain.

Supplementary Text pp. 11–12. These settings are specific to EPIC, not universal NISE defaults.

LASErMPNN designs identities and packing together

Inputs
Backbone residue frames; ligand atoms and elements; optional fixed sequence context.

Representation
Protein–protein and protein–ligand messages on a heterogeneous graph; a pretrained ligand encoder supplies chemical information.

Autoregressive decoding
Choose a random residue order. For each residue, sample amino-acid identity and then its side-chain dihedrals, conditioning on earlier choices.

Why it matters
The next residue sees both what its neighbours are and how their side chains pack.

The network outputs an all-atom sequence design; the independent co-structure predictor tests whether that design reproduces its input.

Fig. 2a; Supplementary Text pp. 5–9, Supplementary Fig. 1.

Training and held-out tests support ligand-aware design

Learning chemical context

Ligand encoder pretraining uses SPICE-2 molecular data to predict charges, dipoles and chemical features. Its weights are frozen during protein-model training.

Protein training uses PDB structures and amino-acid "pseudoligands." Whole backbone frames are idealized and noised to reduce structural artefacts.

Streptavidin fold held out

Binding-site sequence recovery:

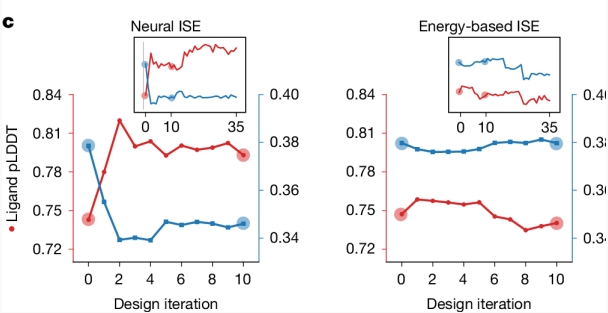
	+ biotin	no biotin
LASErMPNN	53%	38%
LigandMPNN	36%	34%

Boltz-1 self-consistency: 65% versus 13% for the two sequence models. These are

computational recovery tests, not experimental affinities.

Fig. 2; Extended Data Fig. 2; Supplementary Figs. 3–6 and Text pp. 5–13. One held-out fold is not a universal ranking.

Why iterate? Agreement improves along the trajectory



Neural loop: ligand confidence increases while sequence negative log-likelihood (NLL) decreases.

Energy-based loop: using Rosetta minimization and ligand-energy selection does not produce these improvements.

This supports coupled neural refinement for this trajectory. It does not establish that every energy-based protocol fails.

Fig. 1c, cropped; Supplementary Figs. 9–10, 48. Both loops use LASErMPNN for sequence design.

The final hit rate includes substantial downstream filtering

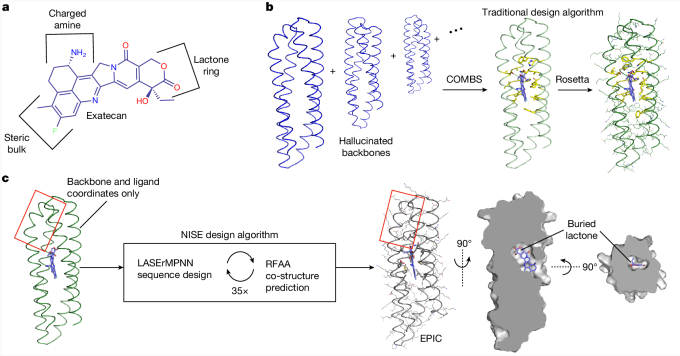
The neural loop optimizes confidence and agreement. Final selection adds:

1. Stricter RMSD/confidence thresholds and packing checks.
2. Relaxation and rejection of ligand clashes in the ligand-free pocket.
3. Ranking by buried, non-hydrogen-bonded polar atoms (BUNs).
4. Predicted dimer/domain-swap checks and visual inspection.

Example: exatecan. Approximately 100,000 pooled sequences led to four experimentally tested proteins. Rosetta is absent from the central NISE optimization loop but is used in final physical assessment. "Neural-only loop" does not describe the entire workflow.

Supplementary Text pp. 11–12, 27–28; Supplementary Figs. 11–12, 31–32.

Test 1: reshape a pocket around exatecan



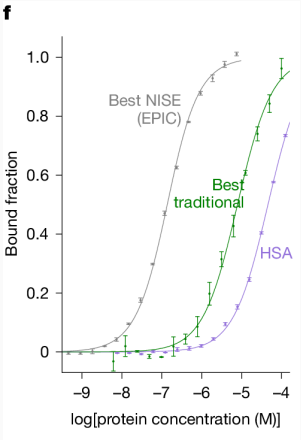
Goal: bury the hydrolysis-sensitive lactone ring.

40 designable four-helix backbones seeded COMBS docking. A top computational COMBS–Rosetta pose became NISE's input.

NISE changes sequence, helix packing and drug burial together.

Fig. 3a–c, cropped; Fig. 3d and Supplementary Fig. 13 describe structural changes.

Exatecan: all four selected designs bind



NISE: 4/4 tested designs bind; $K_d = 0.12$–17 μM.

EPIC: $K_d = 0.12 \pm 0.03$ μM, versus 8 μM for the best initial COMBS–Rosetta binder and 43 μM for albumin.

Initial COMBS–Rosetta panel: 3/16 monomeric binders. Both methods used RFAA quality filters.

Fig. 3f, cropped; Supplementary Fig. 15/Table 1. Plots: mean $\pm$ SD, 3 technical replicates; [ligand] = 50 nM.

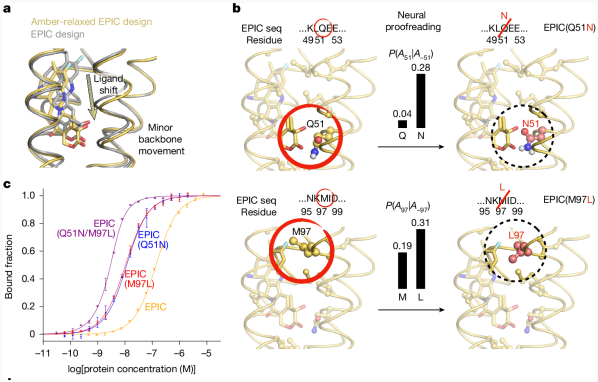
The pocket distinguishes closely related chemistry

Ligand	EPIC K_d	Relative to exatecan
Exatecan	0.12 μM	Intended target
FL118	6 μM	50-fold weaker
Belotecan	19 μM	$\sim$160-fold weaker
Camptothecin	90 μM	$\sim$750-fold weaker

Irinotecan showed negligible binding over the tested range; FITC-linked apixaban and dexamethasone showed no appreciable binding. **Inference:** positive design can create specificity within this panel. No explicit negative-design objective was used, and the panel is not a proteome-wide off-target assay.

Fig. 3g,h; Supplementary Fig. 15 and Table 6. Rounded fold changes calculated from reported K_d values.

Neural proofreading proposes focused improvements



Relax the predicted EPIC complex, then revisit each pocket residue with *all other identities fixed*:

$$P(A_i \mid A_{-i}, B, L).$$

This context differs from the original high-temperature autoregressive design.

Two contact mutations suggested by LASErMPNN were prioritized and tested; no experimental structure was needed to suggest them.

Fig. 4a–c, cropped; Supplementary Fig. 18 and Text p. 14. Amber relaxation occurs before proofreading.

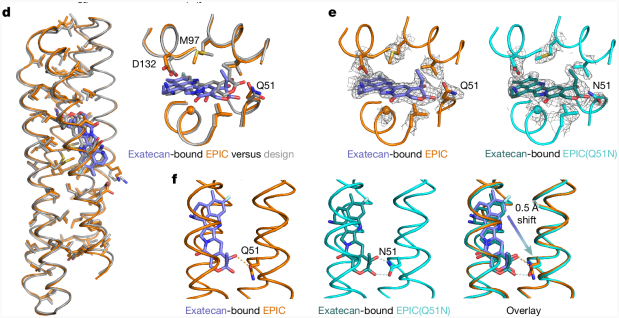
Two substitutions improve affinity by 100-fold

Protein	Exatecan K_d	Gain versus EPIC
EPIC	120 nM	1
Q51N	8.0 ± 1.6 nM	$\sim$15
M97L	7.4 ± 0.7 nM	$\sim$16
Q51N/M97L	1.2 ± 0.2 nM	100

A proposed detrimental control, G104M, bound weakly ($K_d = 57$ μM); variants remained monomeric by size-exclusion chromatography. The paired substitutions support useful local refinement. This is a small targeted mutation test, not a broad validation of NLL as a quantitative affinity predictor.

Fig. 4c; Supplementary Figs. 19–21. Mutant K_d: mean $\pm$ SD across 3 biological replicates.

Crystal structures validate the pocket and explain Q51N



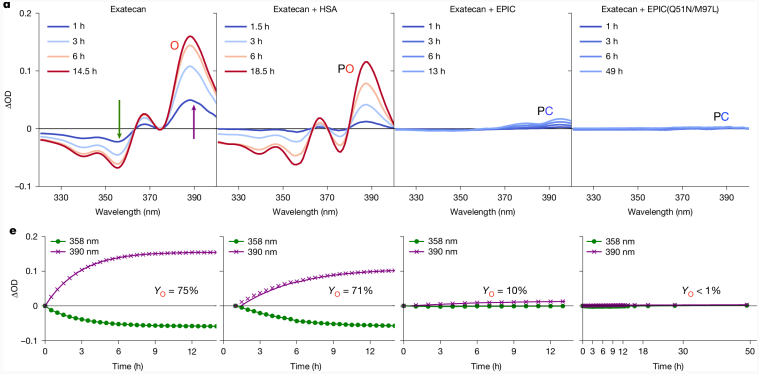
EPIC structure: 2.0 Å resolution; backbone RMSD 0.8 Å to the design input used upstream.

Q51N structure: 2.2 Å. Shorter Asn enables a bidentate hydrogen bond and a ∼0.5 Å deeper ligand position.

RFAA distorted lactone-region geometry. Correct binding mode does not imply exact chemical geometry.

Fig. 4d–f, cropped; Extended Data Figs. 4–5. PDB: 9NZE and 9NZG. APEX has a predicted structure, not a reported crystal structure.

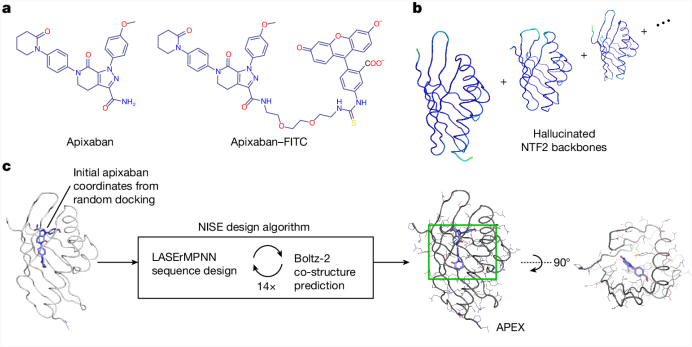
Binding has a functional consequence: protecting the lactone



At room temperature in PBS, pH 7.4: 20 μM exatecan with 20 μM double-mutant protein retains $>$ 99% ring-closed drug for at least 50 h. Albumin alone does not protect it.

Fig. 5d,e, cropped; Supplementary Figs. 24–25. HSA = 500 μM; representative data, at least 2 biological replicates.

Test 2: a new drug and a mixed α/β fold



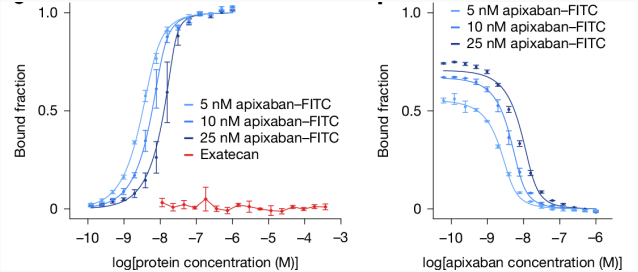
Target: apixaban.
Scaffold: NTF2 pocket.

50 designable representatives derived from 1,615 published backbones; random docking seeds six trajectories.

Boltz-2 replaces RFAA. Optimize ligand pLDDT + $P(\mathrm{bind})$ for 14–28 rounds.

Fig. 6a–c, cropped; Supplementary Text pp. 25–28. Backbones were re-predicted, not coordinate-identical to the prior library.

APEX: competitive binding supports picomolar affinity



5/6 selected proteins bind apixaban with $K_d < 50$ nM.

APEX:

Unmodified apixaban $K_d = 80$ pM (95% CI: 54–122 pM).

Apixaban–FITC $K_d = 990$ pM. Global fits combine direct and competition data; the fluorescent derivative is a distinct ligand.

Fig. 6e,f, cropped; Supplementary Fig. 34. 3 technical replicates; intervals from 1,000 residual-bootstrap fits.

The improvement is large; the denominators need care

Design setting	Binding / tested	Best K_d
Exatecan: NISE	4 / 4	120 nM
Exatecan: COMBS–Rosetta	3 / 16 initial panel	8,000 nM
Apixaban: NISE	5 / 6	0.08 nM
Apixaban: prior LigandMPNN–Rosetta	4 / 9,024	680 nM

These are **experimental hit rates after each workflow's selection**, not the fraction of all computational candidates that bind. Small NISE test panels and different computational budgets limit a universal rate comparison.

Figs. 3, 6 and Discussion; prior apixaban study as reported by Fry et al. Additional same-pose COMBS controls are described separately.

What explains success, and what remains unseparated?

1. **Coupled refinement matters.** Neural selection improves confidence and NLL where the tested energy-based loop does not.
2. **Predictor choice matters.** RFAA would reject the experimentally verified apixaban designs; Boltz-2 enables their selection.
3. **LASErMPNN is not the whole advance.** LigandMPNN inside NISE attains similar aggregate computational metrics, though packing differs.

Missing causal comparison: a matched experimental factorial test of sequence model, structure predictor, iteration and final filters under equal compute.

Fig. 1c; Extended Data Fig. 1; Supplementary Figs. 36–40, 48. Final statement is a journal-club critique.

Limits: confidence is a useful search signal, not an endpoint

1. **Scope:** two drugs and two highly designable fold families. Generalization to arbitrary chemistries and difficult backbones is open.
2. **Calibration:** model confidence correlates with affinity in selected examples; $P(\mathrm{bind})$ does not rank relative affinity of the five apixaban binders.
3. **Application:** drug stabilization is demonstrated in buffer, with albumin controls. Delivery, release, pharmacokinetics and in vivo benefit remain untested here.

A typical 14-round trajectory takes about 5 h on four A6000 GPUs; broad docking and downstream filtering add work beyond that trajectory.

Discussion; Extended Data Fig. 7; Supplementary Text pp. 32–36. Application limits are our interpretation of the reported experiments.

Discussion: which claim would we test next?

Take-home: reciprocal sequence and co-structure models become an optimizer when connected by selection–expansion. EPIC and APEX show that this can yield unusually tight, specific binders with few experimental tests.

1. Does the same protocol work prospectively across a chemically diverse, predeclared target panel?
2. How much gain comes from iteration versus equally expensive one-pass sampling and final filtering?
3. Can confidence-guided negative design improve specificity among closely related ligands without sacrificing affinity?

Prospective questions are our discussion prompts. Code: github.com/polizzilab/LASErMPNN and github.com/polizzilab/NISE.